CyberEscape: Game-based Learning of Cybersecurity Concepts

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CMSC 190 SP 2 Presentation

Background and Problem Statement

With the increase of access to both the Internet and AI-tools for productivity over the past few years, potential risks and threats in cybersecurity have continued to arise through an observed increase in vulnerabilities faced by companies across the country.

Hence, the current state of cybersecurity in the Philippines in terms of awareness and common practices needs further investigation on the technical understanding of underlying mechanisms on well-known cyber attacks to further improve training and programs on cybersecurity and existing safety measures.

Objectives of the Study

1. Design a Game-based learning environment for teaching cybersecurity concepts in a more technical manner
2. Implement a virtual escape room video game to simulate real-life scenarios and applications
3. Incorporate “Capture the Flag” elements to provide a hands-on experience-based learning
4. Assess viability and effectiveness of the video game in deepening awareness of cybersecurity practices & concepts

Significance of the Study

This study provides further insight into **identifying** which areas in cybersecurity Filipino college students **may lack knowledge in**, allowing the development of trainings and programs based on which participants struggled the most with, such as Cryptography.

Furthermore, the significant **increase in post-test scores** provides empirical validation that game-based learning environments, such as virtual escape rooms, are effective platforms for developing in-depth technical skills and understanding of cybersecurity and potential risks.

Scope and Limitations

1. Developed for Windows and Linux
2. Main topics:
   - Web Vulnerabilities
   - Forensics
   - Cryptography
   - Reverse Engineering
3. Free, non-copyright assets and models used
4. Participants: Filipino college students
   - BS CMSC or any other related discipline
   - Any non-CMSC degree

CyberEscape

Method & Materials

Dev Tool

GODOT
Game engine

Godot (v4.6) & GDScript

3D Models and Sounds

KAY KIT™

LowPolyAssets

SOUND JAY

KENNEY

pixabay

Key Features

Four Escape Room Levels

Interactable Puzzles

Answer Randomization

In-Game Hints

Time Measurement

Evaluation

28 Participants

14
CMSC

14
Non-CMSC

Pre- and Post-test

ENJOY Scale

Perceptions Questionnaire

Completion Rate & Time

Pre- and Post-test Scores

Scores of Non-CMSC Respondents

Respondent #	*R-01*	*R-04*	*R-05*	*R-06*	*R-09*	*R-10*	*R-14*	*R-15*	*R-16*	*R-18*	*R-23*	*R-24*	*R-25*	*R-28*
Pre-Test Score	7	11	13	12	10	7	9	6	8	8	1	4	8	5
Post-Test Score	14	20	19	14	18	12	16	8	19	17	19	16	12	14
Gain Score	7	9	6	2	8	5	7	2	11	9	18	12	4	9

Scores of CMSC Respondents

Respondent #	*R-02*	*R-03*	*R-07*	*R-08*	*R-11*	*R-12*	*R-13*	*R-17*	*R-19*	*R-20*	*R-21*	*R-22*	*R-26*	*R-27*
Pre-Test Score	13	12	11	15	13	17	8	18	11	13	10	9	9	12
Post-Test Score	19	18	20	20	18	18	15	20	16	17	20	17	20	18
Gain Score	6	6	9	5	5	1	7	2	5	4	10	8	11	6

Respondents	Pre-Test Score	Post-Test Score	Gain in Score
CMSC	12.2	18.3	6.1
Non-CMSC	7.8	15.6	7.8
Overall (N=28)	10	16.9	6.9

TABLE III

MEAN PRE-TEST AND POST-TEST SCORES

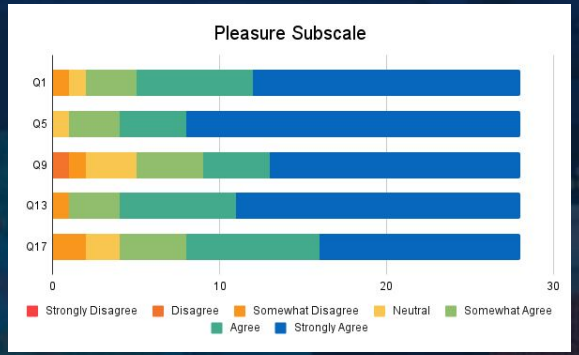
Overall *(n=28)*:

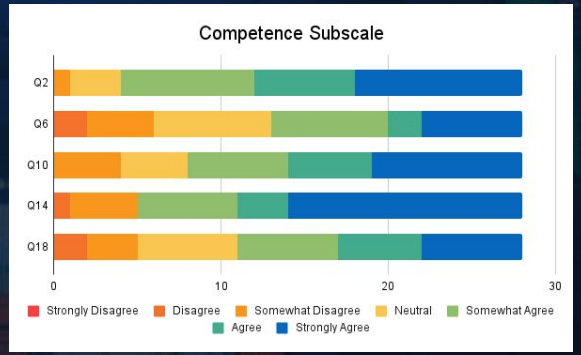
- Paired-samples t-test (n = 28)
- Mean gain in score: **+6.93**
- p-value < 0.001 → **significant difference**

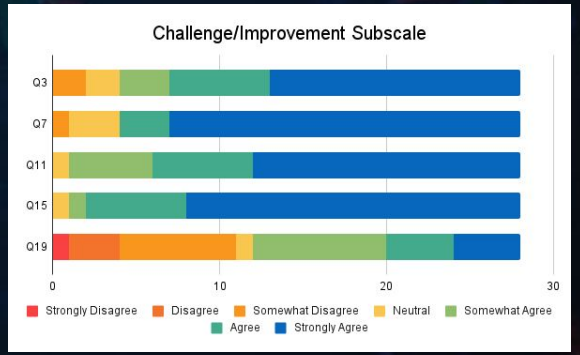
Comparison *(CMSC vs Non-CMSC)*:

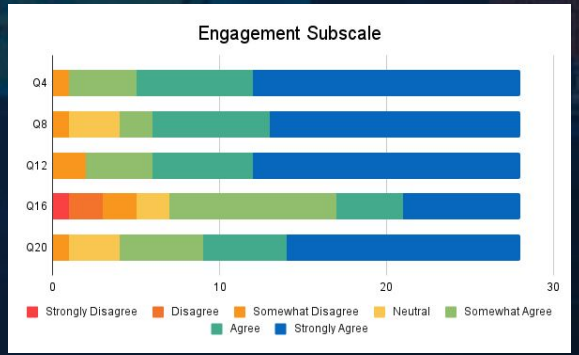
- Independent-samples t-test
- CMSC mean gain: **+6.1**
- NON-CMSC mean gain: **+7.8**
- p = 0.216 → **no significant difference** (both groups benefited equally)

ENJOY Scale









- Measures level of entertainment of participants
- Four subscales each had 5 Qs on a seven-point Likert scale:
  - Pleasure = **6.2143**
  - Competence = **5.3143**
  - Challenge/Improvement = **5.9714**
  - Engagement = **5.95**
- Overall score: **23.45** out of 28 → **Players found the video game entertaining and engaging in learning different cybersecurity concepts**

Effect on Perceptions

#	Question	Before	After
Q1	Programs and data can be made secure so that they cannot be accessed and modified	4.64	4.75
Q2	I am confident in my current safety practices in cybersecurity.	3.25	3.71
Q3	I am confident in the security of the different software and applications that I use.	3.29	3.57
Q4	Interactive learning environments can teach practical skills better than reading static materials.	4.86	4.75
Q5	I find that video games can be effective in learning and developing knowledge.	4.79	4.93

Paired Samples T-Test for Change in Perceptions

Paired Samples T-Test

			Statistic	p		Effect Size
Q1_Before	Q1_After	Wilcoxon W	6.00 [a]	0.374	Rank biserial correlation	−0.429
Q2_Before	Q2_After	Wilcoxon W	32.00 [b]	0.013	Rank biserial correlation	−0.626
Q3_Before	Q3_After	Wilcoxon W	20.50 [d]	0.144	Rank biserial correlation	−0.474
Q4_Before	Q4_After	Wilcoxon W	15.00 [a]	0.374	Rank biserial correlation	0.429
Q5_Before	Q5_After	Wilcoxon W	7.00 [e]	0.240	Rank biserial correlation	−0.500

Note. H_a $\mu_{\text{Measure 1 - Measure 2}} \neq 0$

- Five-point Likert scale questions
- Before playing:
  - Q1: 4.64 *(very high)*
  - Q2: 3.25 *(neutral)*
  - Q3: 3.29 *(neutral)*
  - Q4: 4.86 *(very high)*
  - Q5: 4.79 *(very high)*

- After playing *(Wilcoxon signed-rank test)*:
  - Q2 showed a significant increase **(p-value = 0.013)**: 3.25 → 3.71
  - Game improved confidence in personal cybersecurity practices.

Completion Rate and Time

Level	CMSC	NON-CMSC	Overall (n=28)
Web Vulnerability	1,107.63	976.38	1,041.82
Forensics	681.22	362.1	521.64
Cryptography	1,385.34	1,140.31	1,262.82
Reverse Engineering	990.45	770.01	880.23

TABLE V

AVERAGE COMPLETION TIME PER LEVEL (IN SECONDS)

Level	CMSC	NON-CMSC	Overall (n=28)
Web Vulnerability	85.71%	57.14%	71.43%
Forensics	78.57%	71.43%	75%
Cryptography	57.14%	71.43%	64.29%
Reverse Engineering	57.14%	71.43%	64.29%

TABLE VI

LEVEL COMPLETION RATES PER LEVEL BY GROUP

Avg. Total Time Spent:

- 2,455.82 seconds (~ 40.9 mins.)
- One extreme outlier (13,302s) kept in analysis

Avg. Time per Level:

- Shortest - Forensics (521.64s)
- Longest - Cryptography (1,262.82s)

Completion Rates:

- Highest - Forensics (21/28)
- Lowest - Cryptography and Reverse Engineering (both 18/28)

Time Comparison *(CMSC vs Non-CMSC)*

Group	n	Mean (sec)	Median (sec)	SD	U	p
CMSC	14	2,842	1,709	3,292	93.0	0.836
NON-CMSC	14	2,181	2,172	1,663		

TABLE VII

COMPARISON OF TOTAL TIME BETWEEN CMSC AND NON-CMSC RESPONDENTS

Mann-Whitney U test

p-value = 0.836 → **no significant difference** in total time spent between CMSC and Non-CMSC respondents

This suggests both groups had the same experience going through the levels and/or faced comparable levels of difficulty *(possibly due to differences in prior experience and knowledge)*

Conclusion And Recommendations

Overall, the escape room video game demonstrates promise as an effective, engaging, and viable tool for teaching cybersecurity concepts. Significant pre-test to post-test improvements confirmed its effectiveness, and the ENJOY scale reflected positive reception. Learning gains were consistent across both CMSC and NON-CMSC respondents, and participants' confidence in personal cybersecurity practices also increased significantly.

However, time analysis is not comprehensive yet as not all participants completed every level. Nevertheless, this remains a positive outcome, as knowledge gains were observed even among those who were only able to explore the levels.

Supervised Testing

Alternative Performance Metrics

Pre-Level Briefing Screen

Integrated Hint System

DEMO
SECOND
FIFTH
FOURTH
SIXTH
FIRST
THIRD

РЕКОРД
ВРЕМЯ 464
x2
2k
x1119

Thank you!